

ST JOHNSBURY, VT, USA

WMO: 726140

Lat: 44.420N

Lon: 72.019W

Elev: 212

StdP: 98.80

Time Zone: -5.00 (NAE)

Period: 93-19

WBAN: 54742

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.9	-21.8	-28.5	0.3	-23.2	-25.8	0.4	-21.1	6.8	3.4	5.7	-3.7	0.6	350	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	30.2	21.2	28.6	20.3	27.2	19.4	22.8	27.6	21.8	26.3	20.9	25.0	1.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	16.4	24.7	20.4	15.5	23.8	19.5	14.6	23.0	68.3	27.6	64.8	26.4	61.6	25.2	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.4	4.6	3.7	-29.6	33.2	3.7	1.0	-32.3	33.9	-34.5	34.5	-36.5	35.0	-39.2	35.8		
(5)			-29.8	24.6	3.6	1.0	-32.4	25.3	-34.6	25.9	-36.6	26.5	-39.2	27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-8.2	-7.3	-1.7	6.0	12.7	17.5	20.4	19.3	15.1	8.5	1.9	-4.1
	DBStd	11.18	7.42	6.37	6.00	4.64	4.26	3.62	2.81	3.02	4.02	4.46	4.93	6.35
	HDD10.0	2338	564	485	363	138	18	0	0	0	4	84	245	437
	HDD18.3	4409	822	718	620	371	181	57	12	25	113	305	492	694
	CDD10.0	1159	0	0	2	18	103	227	322	289	156	38	3	1
	CDD18.3	188	0	0	0	0	7	33	76	56	15	1	0	0
	CDH23.3	1662	0	0	2	16	118	342	629	426	124	5	0	0
	CDH26.7	409	0	0	0	2	27	96	163	94	27	0	0	0
Wind	WSAvg	1.0	1.1	1.1	1.3	1.5	1.2	0.9	0.6	0.6	0.7	0.8	1.0	0.9
Precipitation	PrecAvg	954	63	52	63	71	81	94	97	100	82	83	85	78
	PrecMax	1255	173	125	112	158	200	245	184	196	218	218	179	201
	PrecMin	681	10	9	14	15	14	30	30	24	21	10	38	29
	PrecStd	145	36	23	23	30	41	44	36	45	39	46	31	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	10.5	18.7	25.3	30.0	32.3	32.3	31.4	29.9	24.0	17.7	15.3
		MCWB	9.5	6.1	11.1	14.1	18.9	22.0	22.9	22.4	22.1	17.0	13.7	4.2
	2%	DB	7.5	6.5	12.6	21.6	26.6	29.2	30.1	28.9	26.6	20.8	14.5	8.9
		MCWB	3.4	3.4	6.9	12.6	16.7	20.3	21.5	21.0	19.7	14.8	10.9	5.5
	5%	DB	3.9	3.6	9.1	18.0	23.9	27.2	28.5	27.5	24.3	18.1	11.9	6.1
		MCWB	2.1	1.6	4.6	10.7	15.1	18.7	20.5	20.1	18.5	13.5	8.6	3.8
	10%	DB	1.8	1.7	6.6	15.0	21.4	25.1	27.0	25.9	22.3	15.8	9.5	3.7
		MCWB	0.5	0.0	3.4	8.7	14.1	17.8	19.8	19.3	17.5	12.1	6.7	1.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.4	6.9	11.6	17.0	20.2	23.3	24.5	23.6	22.5	19.1	14.8	8.0
		MCDB	12.2	8.8	16.2	22.2	26.8	28.9	29.4	28.2	27.5	21.8	16.7	9.0
	2%	WB	4.6	3.7	7.8	13.9	18.8	21.9	23.0	22.3	21.0	16.6	11.8	5.8
		MCDB	5.9	5.6	11.3	20.0	24.1	27.1	27.6	26.7	24.9	19.1	14.1	8.7
	5%	WB	2.0	1.8	5.4	11.6	17.1	20.6	21.9	21.4	19.8	14.6	9.3	3.7
		MCDB	3.5	3.4	8.6	16.5	21.1	24.8	26.3	25.4	23.1	17.2	11.5	5.9
	10%	WB	0.5	0.3	3.5	9.6	15.4	19.2	21.0	20.5	18.4	12.7	6.9	1.9
		MCDB	2.0	1.5	6.2	14.6	19.7	23.0	25.2	24.1	21.4	15.2	8.8	3.5
Mean Daily Temperature Range	5% DB	MDBR	10.1	11.3	11.1	11.6	12.4	11.7	11.3	11.5	11.4	10.1	8.5	8.2
		MCDBR	10.2	12.7	14.8	18.2	16.9	15.0	13.4	13.0	14.1	13.7	12.2	9.5
		MCWBR	8.7	9.8	9.6	10.2	8.1	6.9	5.9	6.0	7.3	8.5	9.2	7.4
	5% WB	MCDBR	9.8	11.6	13.2	15.8	13.1	12.6	11.1	11.1	11.8	11.4	10.7	8.6
		MCWBR	8.8	9.3	9.3	10.0	6.9	6.5	5.6	5.6	6.8	8.0	9.0	7.5
Clear-Sky Solar Irradiance	taub	0.264	0.283	0.319	0.354	0.374	0.425	0.434	0.406	0.366	0.329	0.308	0.275	
	taud	2.399	2.319	2.339	2.377	2.384	2.238	2.223	2.291	2.398	2.509	2.499	2.441	
	Ebn at Noon	870	910	917	908	896	847	834	846	856	849	809	819	
	Edh at Noon	73	96	108	114	117	136	137	123	102	80	68	65	
All-Sky Solar Radiation	RadAvg	1.43	2.30	3.43	4.27	4.92	5.37	5.47	4.95	4.00	2.37	1.49	1.07	
	RadStd	0.17	0.25	0.22	0.52	0.57	0.48	0.43	0.27	0.29	0.26	0.19	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+63	N/A
(47)	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+70	N/A

Nomenclature: See separate page